Tele Mil: 5016

BY ASIGMA
Military College of Telecommunication Engineering
PIN - 908 768
c/o 56 APO

6625/GS (Tech)/Quantum Secure/T2

14 May 24

DDG IT, DGIS
IHQ of MoD (Army)
New Delhi
PIN - 110 010

WHITELISTING OF 'POST QUANTUM CRYPTOGRAPHY (PQC) BASED WEB SERVER AND BROWSER' BY QUANTUM TECH LAB, MCTE

1. Pl ref SOP on 'Whitelisting of Appl Sw in IA' issued vide DDG IT, DGIS letter No B/04001/Policy/Whitelisting SOP/DDGIT (T&P) dt 19 Jul 22.

2. Military College of Telecom Engg (MCTE) is a premier Cat 'A' Instt of IA imparting trg in contemporary and niche tech domains of Artificial Intelligence (AI), Quantum Tech, 5Gi and EMSO has been recog as a Centre Expertise for Artificial Intelligence (CoE-AI) and in instrumental in infusing AI in the IA.

3. The Post Quantum Cryptography (PQC) Based Web Browser and Server devp as a student proj at MCTE has been selected as one of the best proj in Ideas and innovation Competition 2023-24 (Innoyoddha 2023-24) under the aegis of Army Design Bureau (ADB). Following which, the COAS dir to process the ibid innovation for Intellectual property Right (IPR) and the same has been under prog through ADB.

4. The soln is fwd herewith as per format att as **Appx** duly countersigned by the Dy Comdt & Chief Instructor, MCTE for Whitelisting.

5. For info and necessary action pl.

Anupam

(Anupam Sharma)
Lt Col
GSO-1 (Tech) B
for Comdt

Encls: As above.

Copy to:-

ACG **HQ ARTRAC (IT Sec)**	For info wrt your Letter No B/51106/ArCyGp/T-3/T&E/PQC dt 12 Mar 24
HQ ARTRAC (TT Sec) **DG Sigs (Sigs-1 and Sigs-7)** **CIDSS**	For info pl.

Appx
(Ref Para 4 of letter No 6625/GS (Tech)/Quantum Secure/T2 dt 14 May 24)

STATEMENT OF CASE FOR WHITELISTING OF POST QUANTUM CRYPTOGRAPHY (PQC) BASED WEB SERVER AND BROWSER

INTRODUCTION

1. **Genesis**. Advent of quantum computers will have two fold adverse effect on present day cryptography as under:-

(a) Asymmetric key cryptography algorithms used for initial key negotiations between the server-client communications such as RSA will be vulnerable to quantum computers with adequate capability, rendering the present-day public key cryptography ineffective.

(b) All cryptographic algorithms use an initial random number called 'seed' for generating a secret key for encryption of the communication/message. These seeds are typically generated using Pseudo Random Number Generators (PRNGs). The mathematically generated random numbers using PRNGs can be predicted by Quantum Computers due to their humongous compute power, adding another dimension to vulnerabilities for present day cryptography.

2. **Innovation**. The novel implementation of PQC Based Web Server and Browser based software solution is as under:-

(a) It uses National Institute for Standards and Technology (NIST, USA) recommended PQC based key exchange algorithm Crystal Kyber and digital signature algorithm Falcon. The innovation lies in the usage of both the algorithms at server end for web server and client end for web browser as proof of concept yielding a server client communication that is potentially quantum attack resistant.

(b) All the open-source resources have been utilized for this novel solution. Apache web server, Epiphany web browser, openssl library for cryptographic algorithms and liboqs library for open quantum-safe algorithms have been used to come up with a customized solution of quantum attack-safe server client communications for data networks containing sensitive information.

AIM

3. To develop a PQC Based Web Server and Browser for implementing an encrypted link between server and client at user level against potential quantum computer attack.

OBJECTIVES

4. The PQC Based Web Server and Browser will have the following objectives and features:-

(a) Indigenously developed solution by students and faculty at MCTE.

(b) NIST recommended PQC protocol CRYSTAL Kyber for key exchange.

(c) NIST recommended PQC protocol Falcon for digital signature.

(d) AES-256 (considered to be quantum-attack resistant today as per NIST) for symmetric session key post handshake.

(e) Modified TLS and SSL protocols supporting PQC protocols for establishing a secure session between web server and client.

(f) Integration with hardware QRNG for providing initial high entropy random seed for all cryptographic algorithms.

(g) Customized Apache web server (open source).

(h) Customized Epiphany web browser (open source).

(j) Windows Subsystem for Linux (WSL) for running Epiphany web browser on Windows environment.

Annx I
(Ref Para 9 (a) of DDGIT
Letter No B/04001/Policy/SW
DDGIT (T&P) dt 20 May 2020)

POST QUANTUM CRYPTOGRAPHY (PQC) BASED WEB SERVER AND BROWSER

1. **Short Brief**. Present-day cryptography used for web server-client communication is susceptible to potential quantum computers in near future. To safeguard against this, an innovative hardware-software hybrid solution has been developed by MCTE using NIST recommended PQC protocols and capability for integration with QRNG box. This innovative solutions implements a quantum-attack resistant server-client encrypted communication.

2. **Functional Purpose**. PQC Based Web Server and Browser can safeguard the communication between a web server and a web client (browser) even in the case of adversary gaining access to own networks and is capable of mounting quantum computer attacks on own networks.

3. **Specific Details wrt Sw Proposed to be Devp**.

(a)	SW ID/ Name (Incl ver number).	-	PQC Based Web Server and Browser.
(b)	Est/ Sponsor (Comd) (incl details of PDMG).	-	MCTE, Mhow
(c)	Type of SW (Bespoke/ ERP/ COTS/ Customized).	-	Customized.
(d)	Purposed/ Utility (incl beneficiaries and tgt users).	-	Cyber security in quantum computer threat environment.
(e)	Name of devp org/ vendor name and contact details.	-	MCTE, Mhow.
(f)	Is IPR held with sponsor.	-	Yes (under prog through ADB. Legal drafting stg).
(g)	OS & Sys software reqmts.	-	Win 10/ Win 11/ Linux OS Min 4GB RAM.
(h)	Language/ Platform of SW devp & dply.	-	WSL, Linux.
(j)	Database reqmts (software, ver, ect).	-	Nil.
(k)	Technology dependencies (if any).	-	PQC protocols.
(l)	Cost incl Annual Maint Contract.	-	Nil.
(m)	Architecture to dply (Centralized/ Federated/ Hybrid/ ect).	-	Hybrid.
(n)	Details of Licensing (if any).	-	No.

(o) HW/ Server specifications (if any).	-	i5 processor with 4GB RAM.
(p) Enhancement up gradation.	-	Based on tech adv, MCTE will provide upgrades.
(q) Recommendation, spl instrs and reqmt (if any).	-	Validation agency to be iden for PQC algorithms at a national level.
(r) Intended usage and dply (Standalone/ LAN/ ADN/ Internet).	-	ADN/ LAN/ Internet.

Annx II
(Ref Para 9(a) of DDG IT letter No B/04001/Policy/Sw/ DDG IT (T&P) dt 19 Jul 2022)

CHECKLIST: PRE APLV STG

MANDATORY DETAILS

1.	Name of proj (incl ver).	-	PQC Based Web Server and Browser.
2.	Name of sponsor.	-	MCTE, MHOW.
3.	Type of Sw (Bespoke/ COTS/ Customized).	-	Customized.
4.	Brief justification for devp of Sw appl.	-	Present-day cryptography used for web server-client communication is susceptible to potential quantum computers in near future. To safeguard against this, an innovative hardware-software hybrid solution has been developed by MCTE using NIST recommended PQC protocols and capability for integration with QRNG box. This innovative solutions implements a quantum-attack resistant server-client encrypted communication.
5.	Aim & Scope Purpose incl utility.	-	To develop a PQC Based Web Server and Browser for implementing an encrypted link between server and client at user level against potential quantum computer attack.
6.	To be hosted on internet/ ADN with brief Justification.	-	ADN/ LAN/ Internet as all types of networks will have server client communication that needs to be encrypted against potential quantum computer attacks in near future.
7.	Being devp in house or through IT funds.	-	In-house Devp.
8.	Usability of proposed appls by other arms / services/ org/ est.	-	Pan IA.
9.	Hw and IT infrastructure reqd.	-	Windows 10/Win 11/Linux OS 4GB RAM (min).

10.	Brief details of content of the proposed.	-	Apache server, Epiphany browser, Sw appl.
11.	Endorsement by Head of Br/ Svc/ Fmn.	-	MCTE.
12.	Details of user base.	-	.exe file for usage.

IN PRINCIPLE APPROVAL OF DEPUTY COMMANDANT AND CHIEF INSTRUCTOR FOR DEVELOPMENT OF SOFTWARE 'POST QUANTUM CRYPTOGRAPHY (PQC) BASED WEB SERVER AND BROWSER' BY QUANTUM TECH LAB, MILITARY COLLEGE OF TELECOMMUNICATION ENGINEERING, MHOW

In principle approval for the development of 'Post Quantum Cryptography (PQC) Based Web Server and Browser' is hereby accorded. The software will be used for providing quantum computer attack-resistant web browser and server communication thereby ensuring cyber security post quantum computer era.

Place : Mhow

Date : 14 May 24

MCTE MHOW Date

Maj Gen
Deputy Commandant & CI
Military College of
Telecommunication Engineering